MIL-W-22900A(SHIPS)
15 February 1966
SUPERSEDING
MIL-W-22900(SHIPS)
22 May 1961
(See 6 3)

MILITARY SPECIFICATION

WIND DIRECTION AND WIND SPEED INDICATING EQUIPMENT, TYPE F

1 SCOPE

1 1 This specification covers synchro type F wind direction and wind speed indicating equipment for use on Naval ships.

2 APPLICABLE DOCUMENTS

2 1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein

SPECIFICATIONS

FEDERAL
- FF-B-171 - Bearings, Ball, Annular, General Purpose
- QQ-A-601 - Aluminum—Alloy Sand Castings

MILITARY
- MIL-I-983 - Interior Communication Equipment, Naval Shipboard, Basic Design Requirements for
- MIL-E-17555 - Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
- MIL-S-20708 - Synchros, General Specification for

DRAWINGS

BUREAU OF SHIPS
- 3331250 - Wind Direction and Speed Detector, Type F
- 3331251 - Wind Direction and Speed Transmitter, Type F

BENDIX CORPORATION, FRIEZ INSTRUMENT DIVISION
- 1148595 - Wind Direction and Speed Indicator, Type F

(Copies of specifications standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer)

2 2 Other publications - The following document forms a part of this specification to the extent specified herein Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply

OFFICIAL CLASSIFICATION COMMITTEE
- Uniform Freight Classification Ratings, Rules and Regulations

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd St., New York, N Y 10016)

3 REQUIREMENTS

3 1 Qualification - The wind direction and wind speed indicating equipment furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at time set for opening of bids (see 4 2 and 6 2)

FSC 6660

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3.2 Description - The wind direction and wind speed indicating equipment shall be designed to give continuous and instantaneous visual indications of wind direction in degrees relative to the ship's bow and relative wind speed in knots The equipment shall consist of the following

(a) Wind direction and speed detector
(b) Wind direction and speed transmitter
(c) Wind direction and speed indicator

3 3 General features -

3 3 1 The wind direction and wind speed indicating equipment shall be in accordance with the following paragraphs of MIL-I-983, in addition to the requirements specified herein Whenever a requirement of MIL-I-983 conflicts with a requirement of this specification, the requirement of this specification shall govern

General requirements
Definitions
Materials, general
Substitution of (equal or superior) materials or parts
Unacceptable materials
Acceptable materials
Flammable materials
Arc-resistant materials
Toxic materials
Aluminum
Iron and steel
Nonferrous material (except aluminum)
Zinc
Springs (material)
Other metals
Plastics
Ceramics
Impregnating, embedding and encapsulating compounds
Glass
Lubricants and lubrication
Painting
Protection against corrosion
Bolts, machine screws, studs and nuts
Gaskets
Dials and pointers
Dial sizes
Locking devices
Washers
Ball bearings (precision grade required)

Parts - electrical - general
Capacitors
Variable resistors
Transformers and inductors
Relays
Synchros
Electrical tapes
Dial illumination lamps
Switches
Indicator lights and lampholders
Metallic rectifiers
Enclosures - general
Enclosure - accessibility
Enclosure - degree of
Enclosure - mounting
Stiffening grooves
Minimum sheet metal thicknesses (watertight enclosures)

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Through bolting
Ventilation
Threaded devices
Rounded corners and edges
Internal subassembly protection
Drilled and tapped holes
Welding
Temperature and humidity (special cold limit applies)
Accelerated life
Salt spray
Shock, vibration and inclination (vital vibration test required)
Primary power supply circuits
Power supply tolerances (normal power supply 115 volts 60 cycles single phase a c) (see 3 5 4)
Personnel protection
Shielding and radio frequency noise reduction
Ground potential and grounding
Soldering
Electrical parts mounting
Internal subassembly connection
Terminal boards, connectors and terminals
Synchro connections and markings
Wiring
Dial illumination
Electrical insulation
Dielectric strength and insulation resistance clearances
Airborne noise
Drawings - general
Drawings - preliminary
Drawings - working (see 3 4 5)
Schematic diagrams
Wiring diagram
Assembly drawings
Drawings - manufacturing
Interchangeability and standardization
Manuals
Repair parts (see 3 4 4)
Designation and marking (serial numbers are required)
Reports
Workmanship and general examination

3 3 2 Accuracy and response -

3 3 2 1 The equipment shall be designed so that the error of the wind speed indicator readings shall not exceed the following values

(a) ±1 knot from 1 to 60 knots
(b) ±2 5 knots from 60 to 100 knots
(c) Maximum starting wind speed 1/2 knot

3 3 2 2 The time response of the wind speed transmitter shall be within the limits shown on figure 1

3 3 2 3 The equipment shall be designed so that the error of the wind direction indicator readings shall not exceed ± 2 degrees with wind speed greater than 5 knots

3 3 2 4 The response of the wind direction indication shall be, for any change in wind direction, at the rate of 6 degrees ±1 5 degrees per second

3 3 3 Continuous operation -

3 3 3 1 All wind speed components shall be capable of continuous operation at full scale reading

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3 3 3 2 All wind direction components shall be capable of continuous rotation at a speed of 5 revolutions per minute (r p. m).

3 3 4 Equipment tolerance - Equipment tolerance requirements shall be maintained for nominal voltage and frequency only

3 4 Detail requirements -

3 4 1 Wind direction and speed detector - The detector shall be manufactured in accordance with Drawing 3331250 The detector shall provide a means for generating and transmitting electrical signals, indicative of wind direction and wind speed to the associated transmitter. The detector shall consist of a streamlined vane and a three-blade impeller mounted on a vane support (see figure 2) Any movement of the vane and impeller blades shall be transmitted mechanically to type 18CX4 synchro control transformer in accordance with MIL-S-20708, housed in the vane or vane support The design shall be such that 100 knots of wind speed shall equal 190 r p m of the speed synchro The wind vane and vane support shall be of splashproof construction The exposed nickel-copper alloy shall not be painted, but shall be finished by a sandblast using silica powder - 325 mesh with an air line pressure of approximately 80 pounds per square inch (p s i) and a nozzle size of 1/2-to 3/4-inch diameter The impeller blades shall be molded plastic G E shock resistant rubber phenolic #12809 (or equal). The impeller shall contain a nickel-copper alloy nose cap with captive screw A connector assembly arrangement containing a Pyle National connector, types AP-810 and AR-810, or equal shall be provided to permit removal of the detector mechanism without removing the base from the mounting pintel or disturbing the ship's wiring cable A raincap shall be provided to protect the connector when the detector mechanism is removed The vane tail shall be gray cycolac T-flow as manufactured by Marbon Chemical Company, Division of Borg-Warner (or equal) The center section and forward part shall be monel The connector housing shall contain a 3/4-inch National Pipe thread pipe plug for optional cable connector The design of the detector shall be such as to permit easy disassembling for oiling, cleaning, inspection or replacement

3 4 1 1 Mounting - The detector shall be designed for mounting on the end of a 1-inch diameter vertical pintel The mounting arrangement shall provide a clamping fit and one 5/16-inch through bolt (see figure 2)

3 4 1 2 Balance - The wind vane shall be statically balanced to within 5 ounce-inches of true static balance The impeller shall be statically balanced to within 10 gram-inches

3 4 1 3 Rotor starting torque - With the detector mounted in the normal manner, the rotor shaft shall start to turn at a torque of 0 1 inch ounce

3 4 1 4 Direction vane torque - With the detector mounted in the normal manner with its support horizontal, the vane shall start to rotate at a torque of 4 0 ounce-inches

3 4 1 5 Size - The complete detector assembly shall weigh not more than 21 pounds, and shall not exceed the dimensions shown on figure 2

3 4 1 6 Bearings - All bearings, including those used in the synchros, shall be "sealed-for-life" type, in accordance with FF-B-171

3 4 1 7 Electrical contact, alloy - The composition of the stationary electrical contacts shall be a fine silver graphite alloy, 97 percent silver, 3 percent graphite The material of the rotating contact shall be a silver alloy, 90 percent silver, 10 percent copper Contact pressure shall be 20 $\pm$ 2 grams when tested with a spring scale pull

3 4 2 Wind direction and speed transmitter - The wind transmitter shall be manufactured in accordance with Drawing 3331251 The wind direction and speed transmitter shall contain, in a dripproof enclosure designed for bulkhead mounting, the mechanisms necessary to convert the input from the wind direction and speed detector to relative wind direction in degrees and relative wind speed in knots, and to transmit this intelligence to wind direction and speed indications. The transmitter shall be designed so that it shall be possible to remove and repair either the wind direction or wind speed assembly of the mechanism without disturbing the operation of the other assembly. The speed assembly shall be a servo-amplifier circuit and a roller-disc type tachometer The servo-amplifier shall use a type 18CT4 synchro control transformer as a receiver for signals from the detector representing the rotary speed of the detector rotor assembly. The output signals of the synchro control transformer shall be amplified by a magnetic servo-amplifier circuit

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which shall control the operation of a servomotor This servomotor shall be coupled to a roller-disc type tachometer which angularly positions the rotors of a 400 cycle synchro, type 31TX4, and a 60 cycle synchro, type 37TX6, at a linear rate proportional to the speed of the wind A cutout switch for servomotor shall open the servomotor circuit when wind speed between 95 and 100 knots prevails The direction assembly shall be essentially a servo-amplifier circuit using a type 18CT4, synchro control transformer as a receiver for angular displacement signals representing the position of the vane The output signals of the synchro control transformer are amplified by a magnetic servo-amplifier circuit which operates a servomotor This servomotor mechanically positions, through a gear train, the rotors of a 400 cycle synchro, type 31TX4, and a 60 cycle synchro, type 37TX6, at a predetermined rate (8 ± 1 5 degrees per second) Since angular displacements from the transmitter, in the form of synchro signals, change at an approximately constant rate, wind direction indicators operated by the transmitter provide damped direction values The units shall be bolted in the case and terminal boards shall be provided in the case for incoming wires Two holes fitted with 1-inch National pipe thread pipe plugs shall be provided for attaching stuffing tube fittings Electrical connections shall be made to each unit by plug-jack type terminals, Winchester types A18P and A18S, or equal Power supply tolerances as specified in MIL-I-983 shall apply

3 4 2 1 Sensitivity - The wind direction assembly shall be sufficiently sensitive so that when the wind vane is displaced 8 degrees from a 4 knot wind, it shall upon release, meet the accuracy requirements of 3 3 2 3

3 4 2 2 Size - The transmitter shall be as small and lightweight as practicable but in no case shall dimensions and weight exceed the following exclusive of terminal tubes

Height	-	13-1/2 ±1/8 inches
Width	-	15-1/2 ±1/8 inches
Depth	-	7-3/8 ±1/8 inches
Weight	-	61 pounds

3 4 2 3 Fasteners - Fasteners manufactured from nonferrous and corrosion resisting steel alloys shall be used No carbon or alloy steel fasteners in any form shall be used

3 4 2 4 Enclosure - The base of the enclosure shall be fabricated of cast aluminum alloy conforming to commercial designation 356 T-6, or class 3M, type T-6 of QQ-A-601 The cover shall be fabricated of sheet aluminum alloy Apply one coat of zinc chromate primer followed by two coats of high grade semi-gloss light gray equipment enamel The anodizing, primer and enamel coats shall be applied in accordance with the best commercial practice

3 4 2 5 Power requirement - The power requirements for the transmitter shall be 108 to 123 volts, single phase 60 cycles, 1 5 amperes, 108 to 123 volts, single phase 400 cycles, 4 5 amperes The power requirement for the detector shall be 108 to 123 volts, single phase, 400 cycles

3 4 3 Wind direction and speed indicators - The wind indicator shall be manufactured in accordance with Bendix Corporation Friez Instrument Division Drawing 1148595 The indicators shall contain, in a watertight enclosure (3-foot submergence), the mechanism incorporating types 18TR6 or 18TR4 synchros in accordance with MIL-S-20708 (see 6 1), to receive signals from the wind direction and speed transmitter and to indicate the relative wind direction in degrees and the apparent wind speed in knots The indicator shall contain two synchros on a single chassis Electrical connections shall be made to the case by a plug and jack connector by an Amphenol 26-4100-16P and Amphenol 26-4200-16S, or equal A hole at each end of the case size 1/2 inch National pipe thread shall be provided for cable entrance A power transformer, for the lighting circuit shall be located in the housing A variable resistor for controlling the light intensity shall be provided, and shall be adjusted by a knob from the outside The cover of the housing shall provide a single glass for observing the two dials

3 4 3 1 Dials - The indicator shall have two dials 4 inches in diameter, one for indicating wind direction and the other for indicating wind speed The dials shall be marked in accordance with Drawing 1148595 Red dial illumination shall be provided in the indicator Each dial shall be capable of mounting so that the indicator can be mounted horizontally or vertically by the installing activity.

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3 4 3 2 Size - Indicators shall be as small and lightweight as practicable, but in no case shall dimensions and weight exceed the following

Height - 11-1/8 ±1/8 inches
Width - 7-3/8 ±1/8 inches
Depth - 4-3/16 ±1/8 inches
Weight - 15 pounds

3 4 3 3 The transmitters and indicators provided under this specification shall be equipped with internal terminal boards to facilitate the ship's wiring

3 4 3 4 Fasteners - Fasteners manufactured from non-ferrous and corrosion resisting steel alloys shall be used No carbon or alloy steel fasteners in any form shall be used

3 4 3 5 Enclosure - The enclosure shall be fabricated of cast aluminum alloy conforming to commercial designation 356 T-6, or class 3M type T-6 of QQ-A-601 Apply one coat of zinc chromate primer followed by two coats of high grade semi-gloss light gray equipment enamel The anodizing, primer and enamel coats shall be applied in accordance with the best commercial practice

3 4 3 5 1 Enclosure mounting - Enclosure mounting and arrangements shall be in accordance with Drawing 1148595

3 4 3 5 2 Enclosure accessibility - Enclosure accessibility shall be provided for the ship's wiring arrangements shown on Drawing 1148595

3 4 3 6 Indicator lamps and instrument illumination transformers shall conform to the applicable National Electrical Manufacturer s Association (NEMA) specification requirements

3 4 4 Repair parts - Repair parts and applicable repair parts lists required for maintenance of the parent equipment shall be in accordance with MIL-I-983 The items and quantity of each required for onboard repair parts shall be as shown in table I

Table I - Onboard repair parts required per ship

Item	Quantity	Nomenclature	Bendix Corp part identification number or specification number	Drawing number or type	Drawing piece number
1	1	Pliers "tru-arc"	62562	-	-
2	1	Motor, synchronous	1134619	3331251	220
3	1	Motor, servo	1134617-2	3331251	120
4	1	Motor servo	1134617-1	3331251	232
5	4	Rectifier (1N1591)	1136553-1	3331251	119
6	1	Detector (type F) (less mount	1135924	3331250	-
1/ 7	1	Roller assembly	1134337	3331251	243
1/ 8	1	Synchro	MIL-S-20708	18TRX6	-
1/ 9	1	Synchro	MIL-S-20708	18CX4	-
1/10	1	Synchro	MIL-S-20708	18CT4	-
1/11	1	Synchro	MIL-S-20708	31TX4	-
1/12	1	Synchro	MIL-S-20708	37TX6	-

1/ Not furnished by equipment supplier, shall be obtained from ship s own allowance

3 4 5 Working drawings - In addition to conformance with the requirements of MIL-I-983 working drawings shall give the following information

(a) Power requirements under full load and standby conditions
(b) Heat dissipated under full load and standby conditions
(c) Voltage and resistance readings at key service points
(d) All parts in correct relation as to physical location, with terminals clearly shown and marked and all wiring between parts

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4 QUALITY ASSURANCE PROVISIONS

4 1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements

4 2 Qualification tests[1/] - Qualification tests shall be conducted at a laboratory satisfactory to the Bureau of Ships One complete wind indicating system shall be subjected to the qualification tests specified in table II

Table II - Qualification tests

Test	Requirement paragraph	Test reference
Operating	—	4 5 1
Accuracy test (first part)	3 3 2	4 5 2 1
Supply line voltage and frequency variation	3 3 1	MIL-I-983
Shielding and radio frequency noise reduction	3 3 1	MIL-I-983
Airborne noise	3 3 1	MIL-I-983
Temperature and humidity	3 3 1	MIL-I-983
Inclination	3 3 1	MIL-I-983
Accelerated life	3 3 1	MIL-I-983 and 4 5 3
Accuracy test (second part)	3 3 2	4 5 2 2
Angular displacement retransmission	3 4 2	4 5 4
Sensitivity	3 4 2 1	4 5 4
Dielectric strength and insulation resistance	3 3 1	MIL-I-983
Size	3 4 1 5, 3 4 2 2, and 3 4 3 2	—
Mounting	3 4 1 1	MIL-I983 and 4 5 5
Salt spray (not required for transmitter)	3 3 1	MIL-I-983
Vibration (vital)	3 3 1	MIL-I-983
Shock	3 3 1	MIL-I-983
Accuracy test (third part)	3 3 2	4 5 2 3

4 3 Sampling for quality conformance inspection -

4 3 1 Lot - All equipment offered for delivery at one time shall be considered a lot The lot may include the entire contract quantity or it may be the production of any convenient time period

4 3 2 Group A test - All equipment offered for delivery at one time shall be subjected to the group A test specified in table IV Any equipment failing to meet the requirements of this specification shall not be offered for delivery

4 3 3 Sampling for group B tests - A sample of equipments shall be selected from each lot in accordance with table III and shall be subjected to the group B tests specified in table IV Any equipment failing to meet the requirements of this specification shall not be offered for delivery

1/ Application for qualification tests shall be made in accordance with "Provisions Governing Qualification" (see 6 2 and 6 2 1)

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TABLE III - Sampling for group B tests.

Number of equipments in inspection lot	Number of equipments in sample	(Number of equipments nonconforming on any group B test)	
		Acceptance number	Rejection number
3 and under	All	-	-
4 to 15	3	0	1
16 to 40	5	0	1
41 to 110	7	0	1
111 to 300	10	0	1
301 to 500	15	1	2
501 and over	25	2	3

4 3 4 Sampling for group C tests - Group C tests will be required when the basic design of the equipment or the material of a vital part has been changed One complete equipment shall be subjected to the group C tests specified in table IV The results of each test shall be compared with specification requirements In the event of failure to conform to this specification for any group C test, the contractor shall correct the cause of failure on future production units and repair the deficiency in all equipments produced on the contract or order

4 4 Quality conformance inspection - The sample equipments selected in accordance with 4 3 shall be subjected to the tests specified in table IV The tests shall be performed, in general, in the order listed

Table IV - Quality conformance inspection

Tests	Requirement paragraph	Test reference
Group A		
Operating	-	4 5 1
Group B		
Supply line voltage and frequency variation	3 3 1	MIL-I-983
Dielectric strength and insulation resistance	3 3 1	MIL-I-983
Group C		
Airborne noise	3 3 1	MIL-I-983
Accuracy test (first part)	3 3 2	4 5 2 1
Shielding and radio frequency noise reduction	3 3 1	MIL-I-983
Temperature and humidity	3 3 1	MIL-I-983
Inclination	3 3 1	MIL-I-983
Accelerated life	3 3 1	MIL-I-983 and 4 5 3
Angular displacement retransmission	3 4 2	4 5 4
Sensitivity	3 4. 2. 1	4. 5. 4
Accuracy test (second part)	3. 3. 2	4 5. 2. 2
Size	3.4.1.5,3.4.2.2 and 3. 4. 3. 2	-
Mounting	3 4 1 1	MIL-I-983 and 4. 5 5

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Table IV - Quality conformance inspection, continued.

Tests	Requirement paragraph	Test reference
Group C (cont'd)		
Salt spray (not required for transmitter)	3 3 1	MIL-I-983
Vibration (vital)	3 3 1	MIL-I-983
Shock	3 3 1	MIL-I-983
Accuracy test (third part)	3 3 2	4 5 2 3

4 5 Test procedures -

4 5 1 Operating test - The operating test shall consist of the following for the individual components:

4 5 1.1 Final inspection procedure for wind direction and speed transmitter (detector) - Final inspection procedure for wind direction and speed transmitter (detector) shall be as follows

(a) General examination in accordance with MIL-I-983 and as follows
  (1) Visual examination of unit covering finish identification plate, installation of lock-washers, and similar items
(b) Dielectric strength in accordance with MIL-I-983 and as follows
  (1) Test with Hi-Pot (set at 900 volts) with one lead connected to support assembly and other lead connected successively to each prong of connector for 5 seconds
(c) Insulation resistance in accordance with MIL-I-983 and as follows
  (1) Connect one lead of megger to support assembly and other lead to each prong of connector The resistance shall be not less than 10 megohms at 500 v d c
(d) Direction of rotation and electrical zero position shall be conducted as follows
  (1) Install transmitter on test stand and connect to test indicator Position vane to North position and check test indicator to within plus or minus 1-1/2 degrees Move to each cardinal point and check direction of rotation and accuracy Frictional torque 1 0 inch oz (direction) and 0 1 inch oz (speed) should be verified
  (2) Rotate speed shaft counterclockwise by hand and check direction of indicator
  (3) Adjust each brush to bear on contact slip ring with force of 15 to 20 grams

4 5 1 2 Final inspection procedure for wind direction and speed master transmitter (transmitter) - Final inspection procedure for wind direction and speed master transmitter (transmitter) shall be as follows

(a) Assembled case with terminal strips receptacles and wiring
  (1) General examination in accordance with MIL-I-983 and as follows
    a Visual examination of case assembly covering finish installation of lock-washers, soldered connections, and similar items
  (2) Continuity
    a With Simpson meter (scale RX1) check for continuity between Winchester receptacles and terminal strips
  (3) Dielectric strength in accordance with MIL-I-983 and as follows
    a Position Hi-Pot test plug to a Winchester receptacle Set Hi-Pot to 900 volts and connect one lead to test plug and other lead to spacing collar of Winchester receptacle
    b Test for 5 seconds and then repeat on other connector
  (4) Insulation resistance in accordance with MIL-I-983 and as follows
    a Connect one lead of the megger to the spacing color of the Winchester receptacle Connect the other lead to each terminal of the two terminal strips successively The resistance shall be not less than 10 megohms at 500 v a c
(b) Speed transmitter
  (1) General examination in accordance with MIL-I-983 and as follows
    a Visual examination of speed unit covering finish, installation of lockwashers, soldered connections, and so forth
  (2) Dielectric strength in accordance with MIL-I-983 and as follows
    a Position Hi-Pot test plug to Winchester plug Set Hi-Pot to 900 v and connect one lead to supporting post Test for 5 seconds

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(3) Insulation resistance in accordance with MIL-I-983 and as follows
   a Connect one lead of megger to supporting post of unit Connect other lead successively to each prong of Winchester plug The resistance shall be not less than 10 megohms at 500 v d c
(4) Operational test·
   a Connect unit to test set Energize from line
   b Switch test set to electrical zero The power 31TX4 and 37TX6 synchros should read zero
   c Switch to speed test and check the 37TX6 synchro readings as specified in table V

Table V — Synchro readings

Test set speed (r p m)	Correct intensity knots	Permissible error plus or minus
400	17 5	3/4
800	34 5	3/4
1200	51 0	1
1600	67 5	1-1 4
1800	76 0	1-1/4

Note The values shown under the correct intensity knots apply for nominal voltage (115 volts) and frequencies (60 and 400 cycles) At frequencies other than nominal the values shall be proportional to the change in frequency All values shall be recorded that is
60 cycles 1800 r.p m., 76 knots.
57 cycles 1800 r p m., 76 knots

   d Test set to consist of a 18CX4 synchro control transformer driven by variable speed mechanism with switch for electrical zero position
(c) Direction transmitter in accordance with MIL-I-983 and as follows
(1) General examination
   a Same as (b) (1) a
(2) Same as (b) (2)
(3) Same as (b) (3)
(4) Operational test
   a Connect unit to test set Energize from line
   b Switch test set to electrical zero The 31TX4 and 37TX6 synchro should read zero
   c Switch to direction test and check direction of rotation and accuracy at cardinal points of plus or minus 1-1/2 degrees
   d Test set to consist of 1F synchro with dial and pointer with switch for electrical zero position
(d) Complete assembly
(1) General examination in accordance with MIL-I-983 and as follows
   a Visual examination of finish identification plate, screw and lockwashers

4 5 1 3 Final inspection procedure for wind direction and intensity indicator - Final inspection procedure for wind direction and intensity indicator shall be as follows
(a) Assembled case with terminal strips, receptacles and wiring rheostat and transformer
(1) General examination in accordance with MIL-I-983 and as follows
   a Visual
   b Continuity check of wiring
(2) Dielectric strength in accordance with MIL-I-983 and as follows
   a Hi-Pot at 900 volts for 5 seconds[1/]

[1/] All synchros, transistors and so forth that may be damaged by excessive voltage shall be disconnected during this test

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(3) Insulation resistance in accordance with MIL-I-983 and as follows
a Not less than 10 megohms at 500 v d c
(4) Operation of rheostat

(b) Assembled and wired unit (direction or speed)
(1) General examination in accordance with MIL-I-983 and as follows
a Visual
b Continuity check of wiring
(2) Dielectric strength in accordance with MIL-I-983 and as follows
a Hi-Pot at 900 volts for 5 seconds
(3) Insulation resistance in accordance with MIL-I-983 and as follows
a Not less than 10 megohms at 500 v d c
(4) Accuracy
a Check electrical zero - plus or minus 0 5 degree
b Check direction of rotation
c Check lights

(c) Assembled case and units
(1) Visual check
a Finish screws
b Identification plate

4 5 2 Accuracy tests - Accuracy tests shall be conducted in three parts as follows

4 5 2 1 Accuracy test - first part - An initial accuracy test on the complete wind system shall be conducted in a wind tunnel at the National Bureau of Standards (under auspices of the Naval activity conducting the tests) to determine if the equipment complies with the accuracy requirements specified in 3 3 2

4 5 2 1 1 The complete wind speed system shall be operated at starting and stopping speeds and each wind speed corresponding to 10 knot increments of wind speed over its entire range and shall not show an error in excess of that specified in 3 3 2 1

4 5 2 1 2 The position of the wind vane shall be varied by 30 degrees increments from 0° to 360° and for each position of the vane the wind direction indicators shall meet the requirements specified in 3. 3 2 3.

4 5 2 1 3 From the data obtained in tests specified in 4 5 2 1 1 and 4 5 2 1 2 curves of error versus speed and direction shall be plotted to rectangular coordinates and incorporated in the laboratory report Also a calibration curve of r p m of the impeller versus wind speed shall be included in the report

4 5 2 2 Accuracy test - second part - Upon completion of the first 50 hours of the accelerated life test an accuracy check shall be made to determine if the equipment without adjustment meets all the accuracy requirements as specified in 3 3 2

4 5 2 3 Accuracy test - third part - Upon successful completion of all tests, an accuracy test on the complete wind system shall be conducted in a wind tunnel at the National Bureau of Standards (under auspices of laboratory personnel) to determine that the equipment complies with the accuracy, angular displacement (or transmission and sensitivity requirements as specified in 3 3 2 and 3 4 2 1, respectively

4 5 3 Accelerated life test - During the accelerated life test specified in MIL-I-983 the equipment shall be operated at constantly varying impeller speeds oscillating through the range of 10 to 95 knots at the rate of change of speed of approximately 15 knots per minute except that once each hour the impeller speed shall be steadied at approximately 95 knots for a sufficient length of time to permit determination of indicator error At the same time, the wind vane support shall be oscillated through 120 degrees at the rate of 1/2 cycle per minute to provide a repeated reversal of the follow-up motor in the transmitter Upon completion of the 500 hour test the detector shall be disconnected from the system and with the wind vane locked in place the vane support shall be rotated at a rate of 60 r p m for a period of 4 hours At the end of the 4-hour period, the procedure shall be repeated in the reverse direction for a period of 4 hours

4 5 3 1 Cycling - An equipment cycling test shall be included in the accelerated life test. A start and stop condition that will duplicate actual in-service operation shall be introduced for this phase of the test A minimum of twenty "starts-stops" at the end of each hour's test is required The deenergized period shall be of sufficient duration so as to permit the equipment under test to come to a complete rest The rest period shall not exceed thirty seconds

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4 5 4 Angular displacement retransmission and sensitivity test - Upon completion of the first 50 hours of the accelerated life test, damping and sensitivity tests shall be conducted as follows

4 5 4 1 With the detector operating in a normal position, set the wind vane to indicate a definite direction and de-energize the system While de-energized, rotate the wind vane 120 degrees, and with the vane in this position energize the system noting the time required for the wind direction indicator to become synchronized The time required to reach synchronization, within ± 1 5 degrees, shall be between 12 6 and 18 4 seconds

4 5 4 2 With the detector in a position so that the plane surface of the wind vane is horizontal, verification of the requirement of 3 4 1 3 shall be made

4 5 4 3 Maximum frictional torque of detector rotor shaft shall equal that of starting wind speed of 2 knots

4 5 5 Mounting - Compliance with the requirement of 3 4 1 1 through 3 4 1 5, as well as special features of the various component mountings shall be determined in general accordance with MIL-I-983

4 5 6 During the entire accelerated life, shock, and vibration tests, no indicator shall show an error in excess to that specified in 3 3 2

5 PREPARATION FOR DELIVERY

5 1 Domestic shipment and early equipment installation and for storage of onboard repair parts -

5 1 1 Wind direction and wind speed indicating equipment -

5 1 1 1 Preservation and packaging - Preservation and packaging which may be the supplier's commercial practice shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation

5 1 1 2 Packing - Packing shall be accomplished in a manner which will insure acceptance by common carrier at the lowest rate and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation The shipping containers or method of packing shall conform to the Uniform Freight Classification Ratings, Rules and Regulations or other carrier regulations as applicable to the mode of transportation and may conform to the supplier's commercial practice

5 1 1 3 Marking - Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor s commercial practice The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, manufacturer's name and destination

5 1 2 Onboard repair parts - The onboard repair parts shall be preserved and packaged level A, packed level C and marked levels A and C respectively, in accordance with MIL-E-17555

5 2 Domestic shipment and storage or overseas shipment -

(5 2 1 The following provides various levels of protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made (see 6 1)

5 2 1 1 Preservation packaging, packing and marking - The equipment and accessories, repair parts and technical publications shall be preserved and packaged levels A or C packed levels A or B, and marked in accordance with MIL-E-17555)

6 NOTES

6 1 Ordering data - Procurement documents should specify the following

(a) Title, number, and date of this specification
(b) Quantity of items required
(c) Quantity of manuals required (see 3. 3 1).
(d) Whether indicators require types 18TR6 or 18TR4 synchros (see 3 4 3)

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(e) Whether Federal stock number is to be entered on identification plate
(f) Inventory control point for repair parts
(g) Preservation, packaging, packing or marking requirements other than those required by 5 1 (see 5 2)

6 2 With respect to products requiring qualification awards will be made only for products which are at the time set for opening of bids qualified for inclusion in Qualified Products List QPL-22900 whether or not such products have actually been so listed by that date The attention of the suppliers is called to this requirement and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for products covered by this specification The activity responsible for the Qualified Products List is the Bureau of Ships Department of the Navy Washington D C 20360 and information pertaining to qualification of products may be obtained from that activity Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification" (see 6 2 1)

6 2 1 Copies of "Provisions Governing Qualification" may be obtained upon application to Commanding Officer Naval Supply Depot 5801 Tabor Avenue Philadelphia Pa 19120

6 3 CHANGES FROM PREVIOUS ISSUE THE EXTENT OF CHANGES (DELETIONS ADDITIONS ETC) PRECLUDE THE ANNOTATION OF THE INDIVIDUAL CHANGES FROM THE PREVIOUS ISSUE OF THIS DOCUMENT

Preparing activity
Navy - SH
(Project 6660-N011SH)

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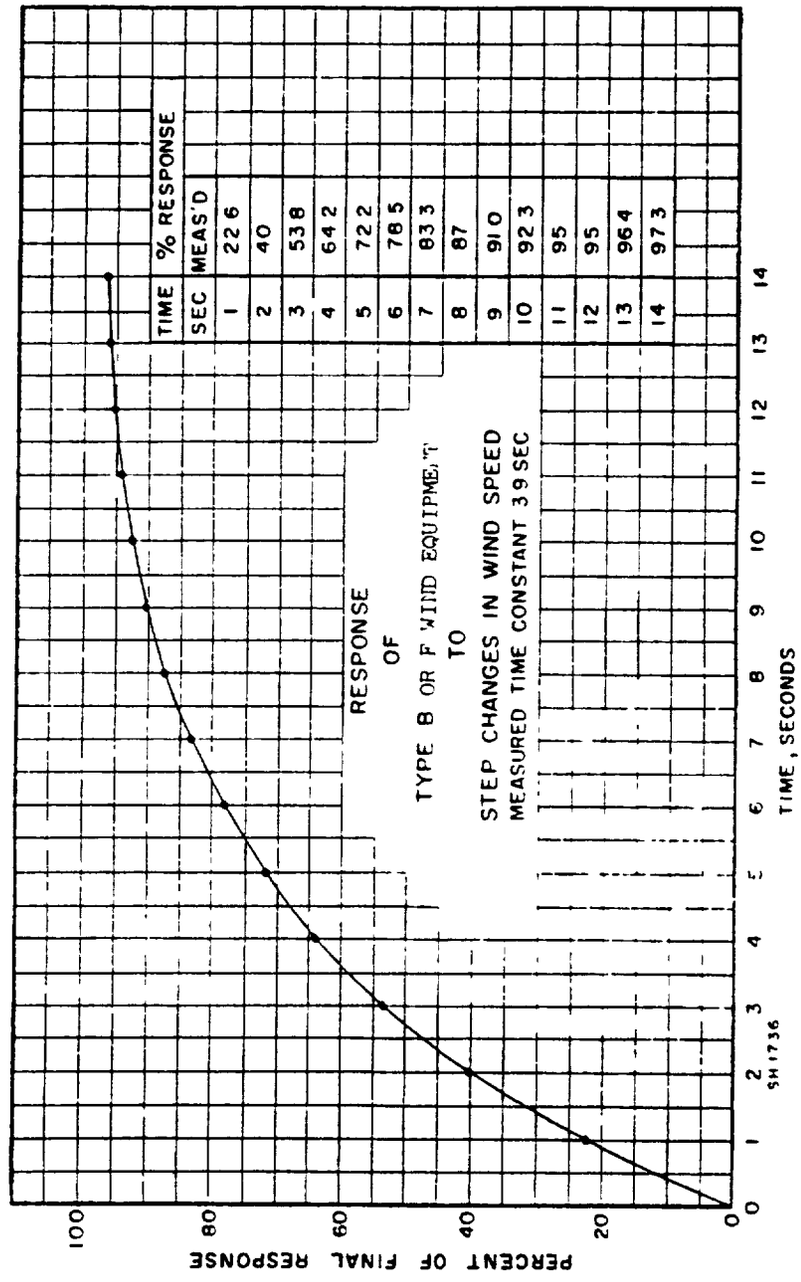


FIGURE 1 - Response of wind equipment

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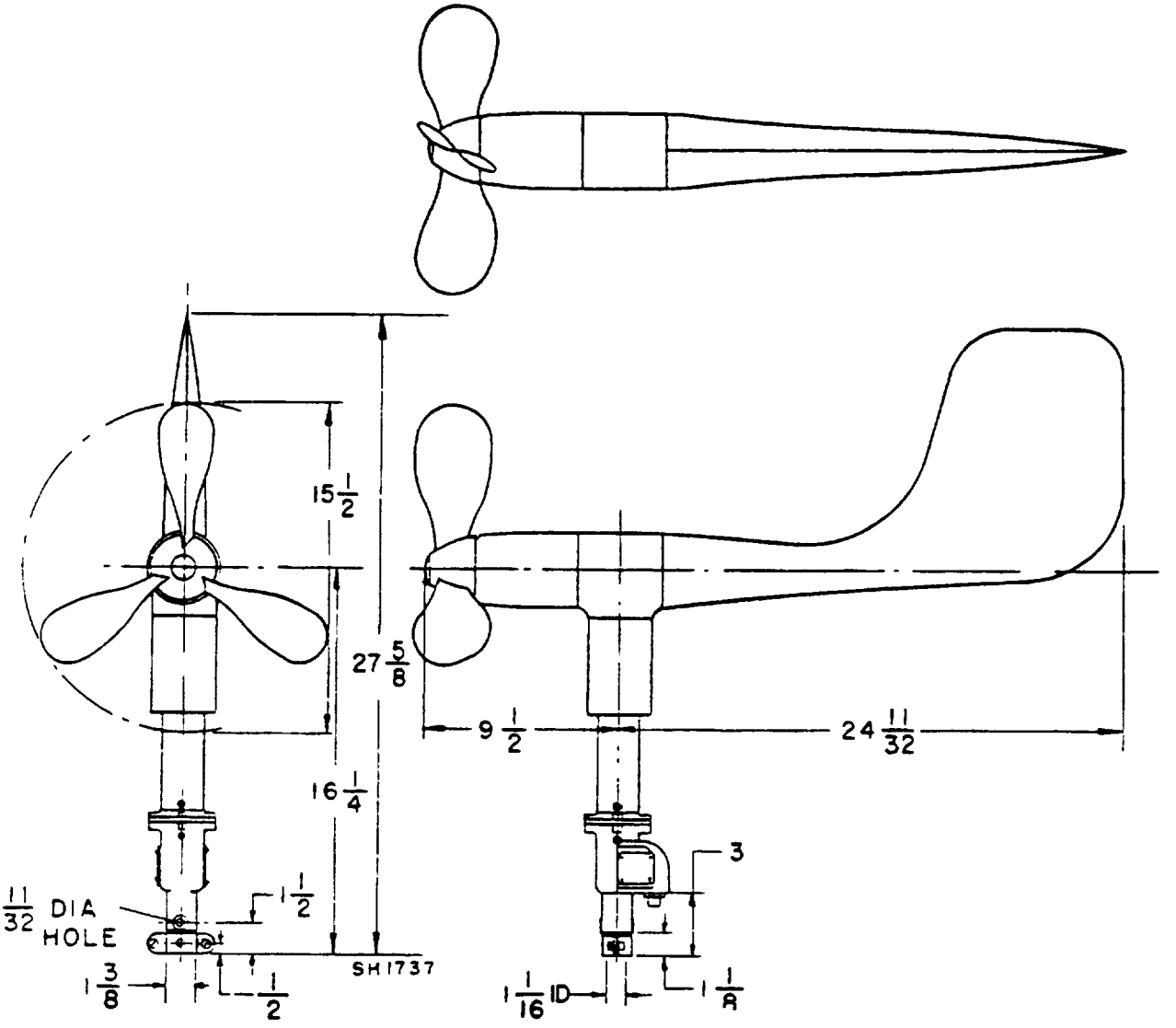


Figure 2 - Dimensions for detecting assembly

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost Comments and the return of this form will be appreciated Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD, FORM 1 APR 84 1426

REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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